

Total No. of Questions : 6]
P5040

SEAT No. :
[Total No. of Pages : 2

T.E./Insem. - 538
T.E. (Computer Engineering)
DATA COMMUNICATION AND WIRELESS SENSOR
NETWORKS
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Solve Que.1 or 2, Que.3 or 4, Que.5 or 6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data if necessary.*
- 4) *Figures to right indicate full marks.*

- Q1)** a) Encode binary data stream 11001110 using following line coding schemes : **[6]**
- i) Manchester code
 - ii) Differential Manchester
 - iii) Polar (RZ, NRZ)
- b) Write a note on Bluetooth frame format. **[4]**

OR

- Q2)** a) Explain different sampling techniques with diagram & example. **[6]**
- b) What is sampling Frequency, Nyquist frequency, Sampling rate & Nyquist Rate? **[4]**
- Q3)** a) Explain sliding window protocol with an example. **[6]**
- b) Explain Circuit switching with suitable diagram. **[4]**

OR

P.T.O.

- Q4)** a) Compare circuit switching & packet switching techniques. [4]
b) Write in detail the working of CSMA/CA and detail the flow diagram.[6]

- Q5)** a) Draw and explain the architecture of wireless sensor network. [6]
b) What is RFID? List out the differences between RFID and Barcode.[4]

OR

- Q6)** a) Explain the use of WSN in Military application & Robots. [4]
b) Explain category 1 and category 2 sensor networks with features. [6]

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